Southern New Hampshire University

CS 340 README

About the Project/Project Title

This project implements a dashboard that's interactive and is meant to visualize animal shelter data using Python, MongoDB, and Dash. The dashboard allows users to filter animals by outcome type, and the view is detailed and shows information in an interactive data table that visualizes top breeds with a pie chart and explores geolocation data via a map.

Motivation

The motivation for this project is to create a simplistic, clean, user-friendly interface for exploring our animal shelter data. By combining visuals as well as filters and geolocation maps, the dashboard helps shelter staff and others analyze trends and outcomes in their animal population.

Getting Started

This is an example of how you may give instructions on setting up your project locally: "To get a local copy up and running, follow these simple example steps."

Installation

To run this project, Jupyter Notebook will be needed, and ensure Python 3.9

Install the required Python libraries using pip: pip install pandas numpy matplotlib plotly dash dash-leaflet jupyter-dash pymongo

Ensure that the MongoDB database is also accessible with proper user name and password credentials.

Usage

In order to start launching JupyterLab or Jupyter Notebook

Open your ipynb file

Run all cells and a JupyterDash server

Use the radio buttons to filter animals by outcome type

Click rows in the data table to update the map to show the selected animals' location

Observe the pie chart updates reflecting the top breeds in the current filter

Code Example

@app.callback(Output('datatable-id','data'), [Input('filter-type', 'value')]) def update_dashboard(filter_type): query = {} # default: show all if filter_type.lower() == 'adoption': query = {'outcome_type': 'Adoption'} df_filtered = pd.DataFrame.from_records(db.read(query)) if '_id' in df_filtered.columns: df_filtered.drop(columns=['_id'], inplace=True) return df_filtered.to_dict('records')

This callback filters the table dynamically based on the user's selection

Tests

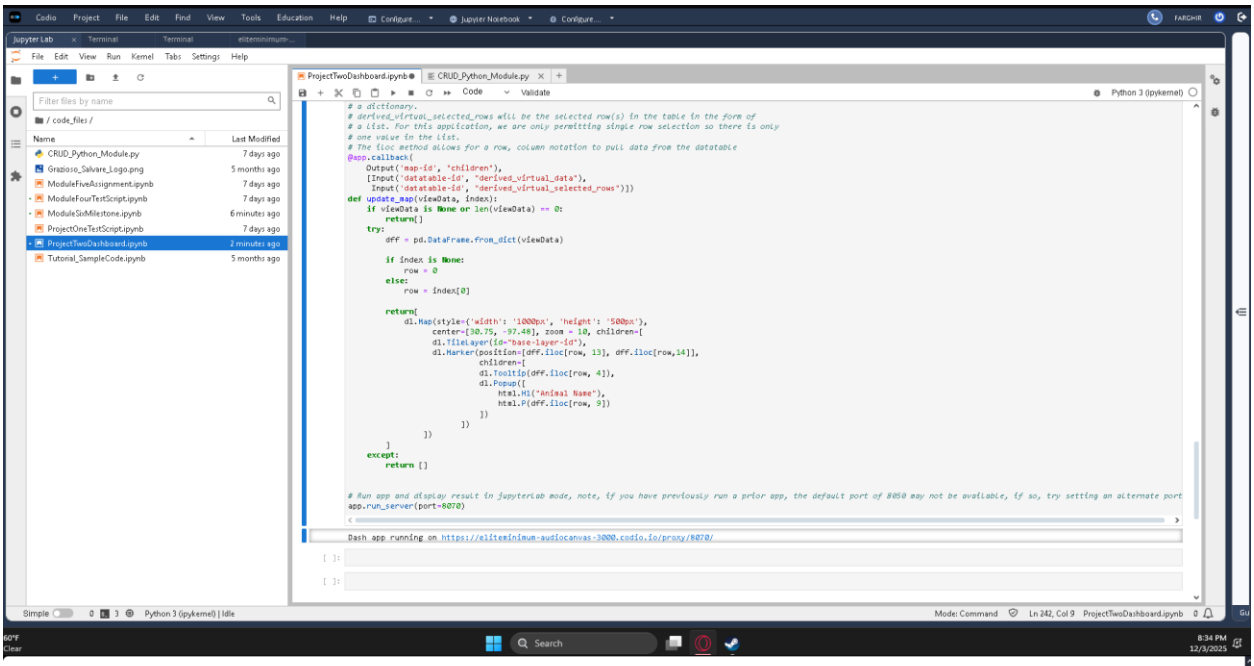
The dashboard is interactive and can be tested by selecting different outcome types and verifying table updates.

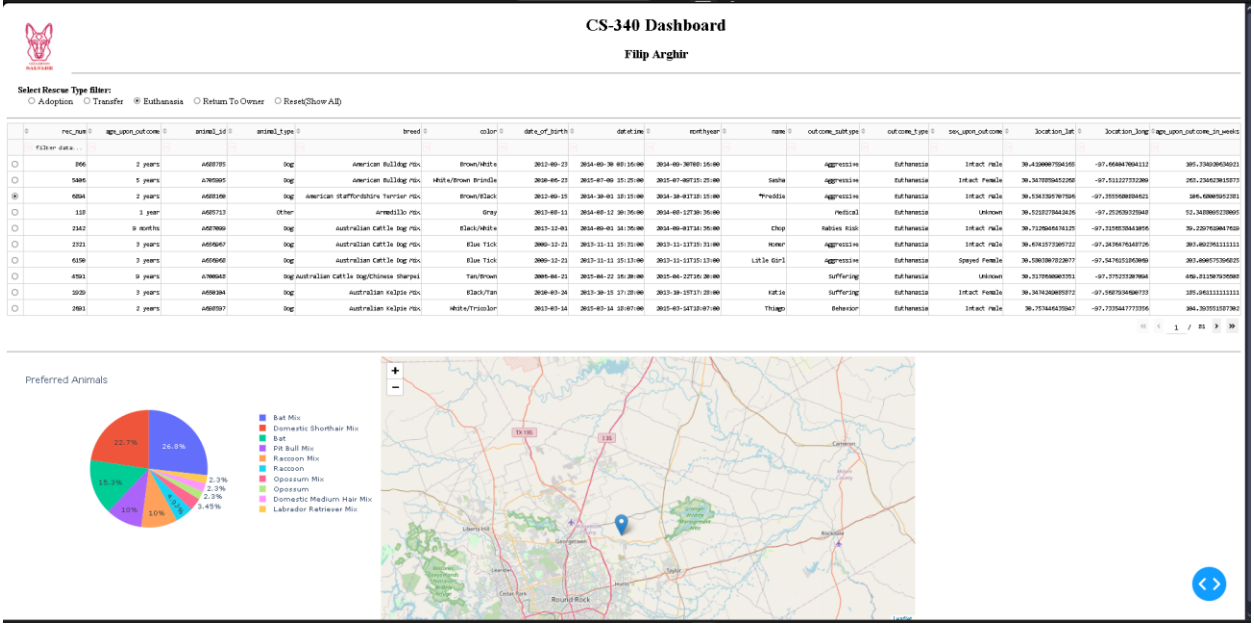
Clicking rows in the table and verifying that the map updates correctly

Confirming that the ie chart reflects the top 10 breeds dynamically

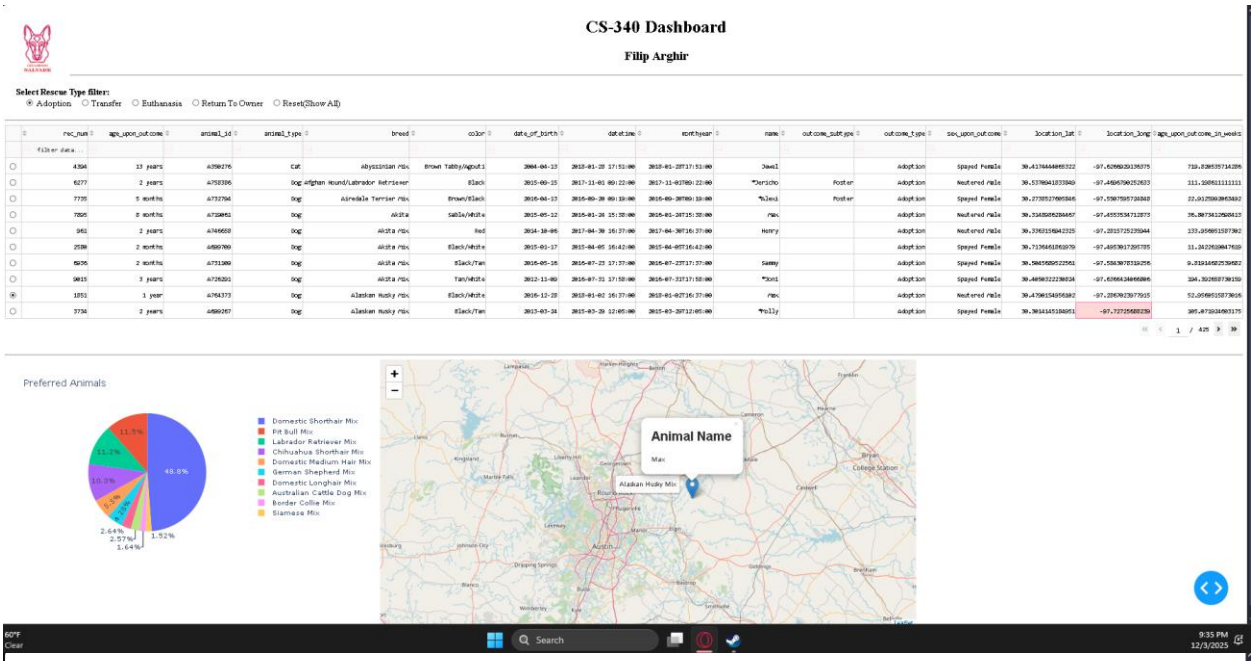
Screenshots

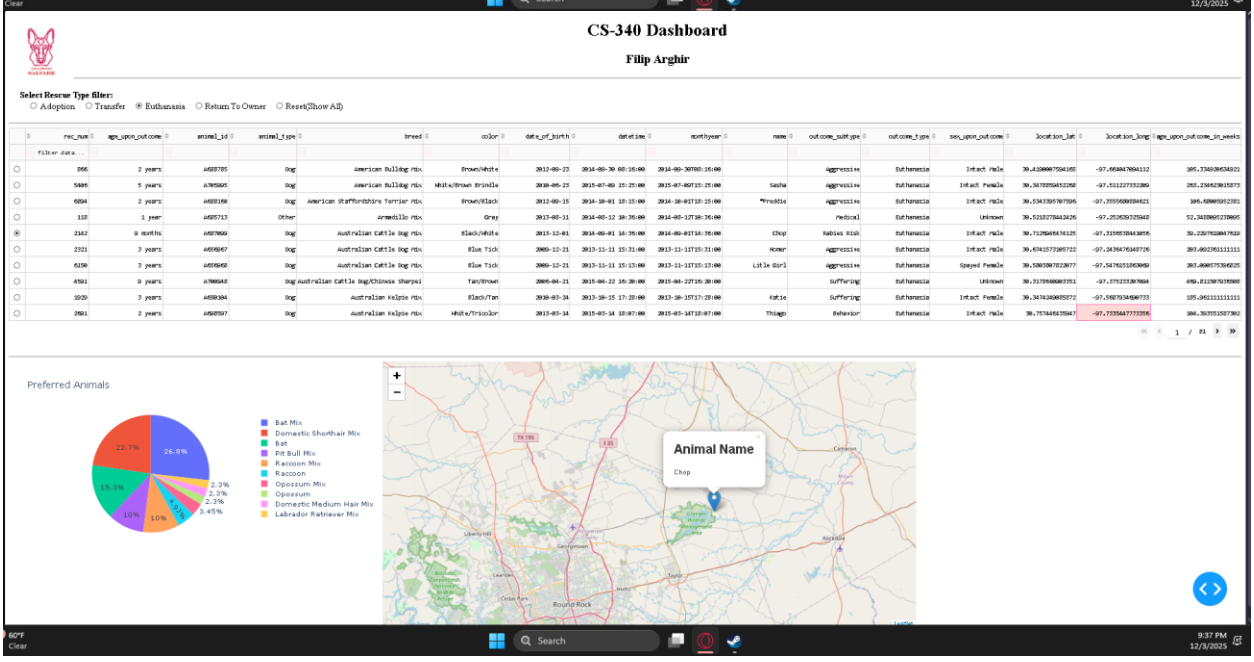
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CS-340 Dashboard

Filip Arghir

Select Rescue Type filter:

○ Adoption ◉ Transfer ○ Euthanasia ○ Return to Owner ○ Reset(Show All)

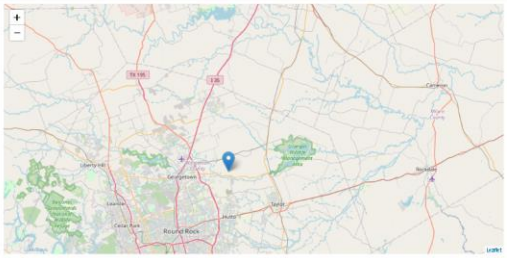
	rec_num	age_upon_outcome	animal_id	animal_type	breed	color	date_of_birth	datetime	monthyear	name	outcome_subtype	outcome_type	sex_upon_outcome	location_lat	location_long	age_upon_outcome_in_weeks
	filter data...															
○	5253	1 year	A743237	Dog	Chihuahua Shorthair mix	Brown/White	2015-02-04	2017-02-09 14:55:00	2017-02-09T14:55:00		Partner	Transfer	Spayed Female	30.616706881741 6	-97.7355585853982	52.4887856825397
○	5276	3 months	A737883	Dog	Chihuahua Shorthair mix	Brown/Tan	2016-07-30	2016-10-30 12:45:00	2016-10-30T12:45:00		Partner	Transfer	Intact Female	30.675539765 4361	-97.6661259270227	13.21875
○	5372	11 years	A627488	Dog	Chihuahua Shorthair mix	Buff	2005-01-14	2016-01-15 15:07:00	2016-01-15T15:07:00	Buster	Partner	Transfer	Neutered Male	30.561181857 6768	-97.5187581837302	574.518551587302
○	5387	1 year	A763917	Dog	Chihuahua Shorthair mix	Tan/Tan	2016-12-10	2017-12-21 13:14:00	2017-12-21T13:14:00	Flor	Partner	Transfer	Spayed Female	30.365554253 2366	-97.7331568830368	52.6561084126984
○	5452	1 day	A768888	Dog	Chihuahua Shorthair mix	Black/White	2016-12-11	2016-12-12 08:00:00	2016-12-12T08:00:00	*Hamston	Partner	Transfer	Intact Male	30.325861139 6547	-97.6180207183567	0.1428571428571 43
○	5477	2 years	A734225	Dog	Chihuahua Shorthair mix	White/Chocolate	2013-10-19	2015-10-23 15:22:00	2015-10-23T15:22:00	*Kiki	Partner	Transfer	Intact Female	30.7486736662588	-97.5385763285535	104.948611111111
○	5516	4 months	A762383	Dog	Chihuahua Shorthair mix	Tan	2017-07-17	2017-11-10 09:13:00	2017-11-10T09:13:00		Partner	Transfer	Intact Female	30.7145480684454	-97.7826543841 98	17.913863568254
○	5765	1 month	A765564	Dog	Chihuahua Shorthair mix	Black/Brown	2017-12-15	2018-01-24 00:00:00	2018-01-24T00:00:00	*Jasper	Partner	Transfer	Intact Male	30.6472185945935	-97.7540176803634	5.7142857142857 1
○	5826	1 year	A713571	Dog	Chihuahua Shorthair mix	Brown Brindle/White	2014-10-08	2015-10-13 11:30:00	2015-10-13T11:30:00	*Deedee	Partner	Transfer	Intact Female	30.3954364163456	-97.7434715638669	52.026488095 2381
○	5938	6 years	A753534	Dog	Chihuahua Shorthair mix	Tan	2011-07-07	2017-07-14 12:27:00	2017-07-14T12:27:00	*Valoria	Partner	Transfer	Spayed Female	30.6611856631385	-97.6282464777363	314.218464285714

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Preferred Animals



Domestic Shorthair Mix
Chihuahua Shorthair Mix
Pit Bull Mix
Domestic Medium Hair Mix
Labrador Retriever Mix
Domestic Longhair Mix
German Shepherd Mix
Siamese Mix
Dachshund Mix
Australian Cattle Dog Mix



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CS-340 Dashboard

Filip Arghir

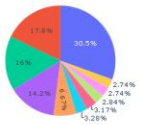
Select Rescue Type filter:

○ Adoption ○ Transfer ○ Euthanasia ◉ Return to Owner ○ Reset(Show All)

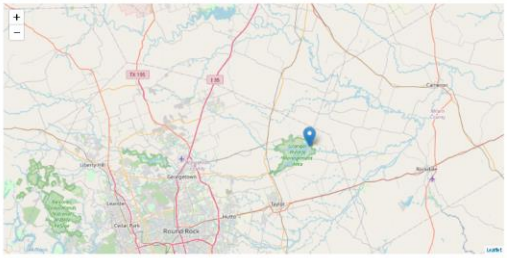
	rec_num	age_upon_outcome	animal_id	animal_type	breed	color	date_of_birth	datetime	monthyear	name	outcome_subtype	outcome_type	sex_upon_outcome	location_lat	location_long	age_upon_outcome_in_weeks
	filter data...															
○	7399	3 years	A678921	Dog	Chihuahua Longhair/Cairn Terrier	Tan/Brown	2011-01-15	2014-01-16 13:56:00	2014-01-16T13:56:00	Zoe		Return to Owner	Spayed Female	30.7496917967782	-97.4844343453825	156.654365079365
○	2578	2 years	A753306	Dog	Chihuahua Longhair/Dachshund Hybrid	Black	2015-07-05	2017-07-18 16:59:00	2017-07-18T16:59:00	L.D.		Return to Owner	Intact Male	30.2759198158711	-97.4724085867416	104.380865555556
○	1892	1 year	A692642	Dog	Chihuahua Shorthair	Tan/White	2013-11-23	2015-08-27 18:33:00	2015-08-27T18:33:00	Millie		Return to Owner	Spayed Female	30.6459459679969	-97.5709378788341	91.8367233869534
○	2691	2 months	A714458	Dog	Chihuahua Shorthair	Blue/White	2015-08-23	2015-10-24 16:41:00	2015-10-24T16:41:00	Rex		Return to Owner	Neutered Male	30.2787181913487	-97.3568557815789	8.9566434112588 41
○	3762	1 year	A692642	Dog	Chihuahua Shorthair	Tan/White	2013-11-23	2014-11-28 15:54:00	2014-11-28T15:54:00	Millie		Return to Owner	Intact Female	30.7271654851539	-97.5965733858185	52.0478174603175
○	4484	2 years	A761796	Dog	Chihuahua Shorthair	Black	2015-11-09	2017-11-20 17:53:00	2017-11-20T17:53:00	Mini		Return to Owner	Spayed Female	30.3259018388635	-97.6170835668 43	104.10625
○	5165	13 years	A659585	Dog	Chihuahua Shorthair	Tan/White	2002-10-15	2015-11-01 16:43:00	2015-11-01T16:43:00	Juan		Return to Owner	Neutered Male	30.3566285821272	-97.6926280365377	680.858277777778
○	27	2 years	A764418	Dog	Chihuahua Shorthair mix	Tan/White	2015-12-30	2017-12-30 11:58:00	2017-12-30T11:58:00	Burro		Return to Owner	Neutered Male	30.2771625844418	-97.7122755987492	104.628537369 59
○	72	1 year	A712292	Dog	Chihuahua Shorthair mix	Tan/White	2014-09-20	2015-09-21 12:10:00	2015-09-21T12:10:00	Peanut		Return to Owner	Spayed Female	30.7565508876433	-97.6364223787824	52.3592857142856
○	86	6 years	A730822	Dog	Chihuahua Shorthair mix	Tan/Red	2011-04-10	2017-05-24 16:15:00	2017-05-24T16:15:00	Pepe		Return to Owner	Neutered Male	30.4854162362655	-97.5934475588179	310.525357619048

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Preferred Animals

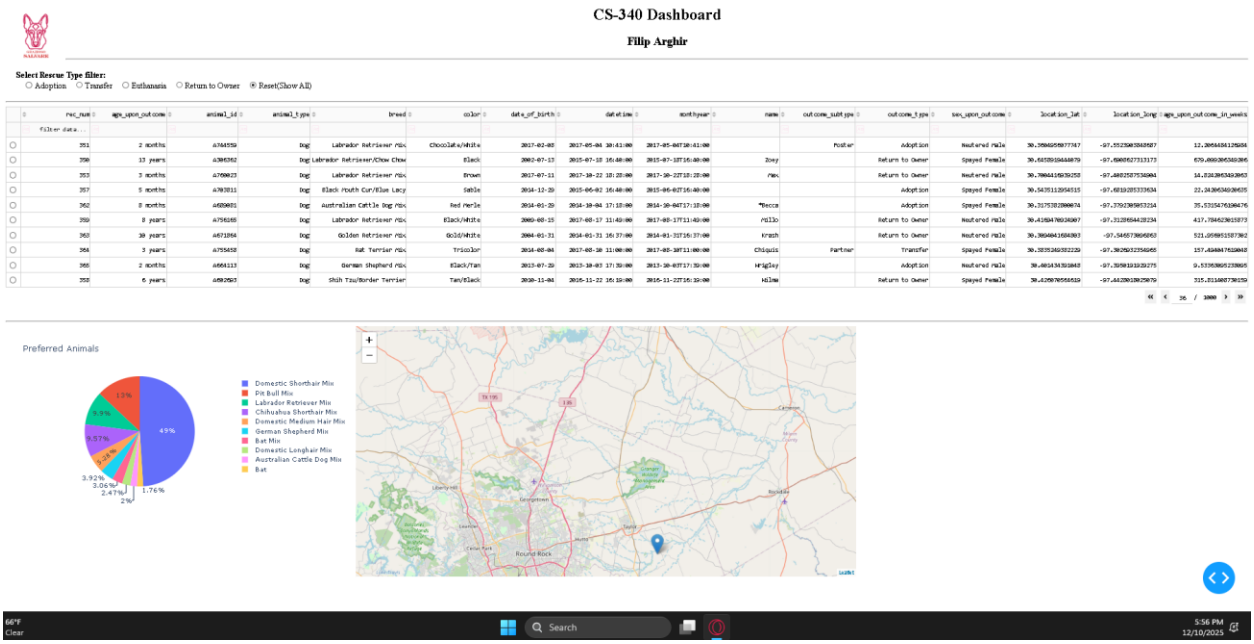


Pit Bull Mix
Labrador Retriever Mix
Chihuahua Shorthair Mix
Domestic Shorthair Mix
German Shepherd Mix
Miniature Poodle Mix
Siberian Husky Mix
Dachshund Mix
Australian Cattle Dog Mix
Boxer Mix



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Contact

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